

GED Math Worksheet: Polynomials, Factoring & Quadratic Equations

Name: _____

Date: _____

Instructions: Answer all 40 questions. Show your work where applicable. Use separate paper if needed.

SECTION 1: POLYNOMIAL OPERATIONS (Questions 1-12)

Adding and Subtracting Polynomials

1. Simplify: $(3x^2 + 5x - 2) + (2x^2 - 3x + 4)$
2. Simplify: $(7y^2 + 4y - 1) - (3y^2 + 2y - 5)$
3. Simplify: $(4a^3 + 2a^2 - a) + (a^3 - 3a^2 + 2a)$
4. Simplify: $(6b^2 - 5b + 3) - (2b^2 + b - 7)$
5. Combine like terms: $5x^2 + 3x - 2 + 4x^2 - x + 6$

Multiplying by a Monomial

6. Multiply: $3x(2x^2 + 4x - 5)$
7. Multiply: $-2a(3a^2 - 5a + 1)$
8. Multiply: $4xy(2x^2 + 3xy - y^2)$
9. Multiply: $5m^2(m^3 + 2m - 3)$

Multiplying Binomials (FOIL)

10. Multiply: $(x + 3)(x + 8)$
11. Multiply: $(2y - 1)(y - 6)$

12. Multiply: $(a + 4)(a - 7)$
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SECTION 2: FACTORING (Questions 13-26)

Factoring Out the GCF

13. Factor: $6x^2 + 18x + 12$
14. Factor: $24x^4 + 48x^3 + 56x^2$
15. Factor: $15a^3b + 25a^2b^2 - 10ab$
16. Factor: $12m^2 - 8m + 4$

Factoring Trinomials ($a = 1$)

17. Factor: $x^2 + 4x - 12$
18. Factor: $x^2 + 11x + 24$
19. Factor: $y^2 - 7y + 12$
20. Factor: $z^2 - 2z - 15$

Factoring Trinomials ($a > 1$)

21. Factor: $2x^2 + 9x + 4$
22. Factor: $3y^2 + 10y + 3$
23. Factor: $4a^2 - 11a + 6$

Special Factoring Patterns

24. Factor: $x^2 - 49$ (Difference of Squares)
25. Factor: $x^2 + 6x + 9$ (Perfect Square Trinomial)
26. Factor: $4a^2 - 20a + 25$ (Perfect Square Trinomial)
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SECTION 3: QUADRATIC EQUATIONS (Questions 27-40)

Solving by Factoring

- 27. Solve: $x^2 - 2x = 15$
- 28. Solve: $m^2 - 8m + 16 = 0$
- 29. Solve: $y^2 + 5y - 24 = 0$
- 30. Solve: $2x^2 - 7x + 3 = 0$

Solving by Completing the Square

- 31. Solve: $x^2 - 12x + 20 = 7$
- 32. Solve: $x^2 + 6x - 16 = 0$
- 33. Solve: $a^2 - 8a + 12 = 0$

Solving Using the Quadratic Formula

- 34. Solve: $3x^2 + 11x - 4 = 0$
- 35. Solve: $m^2 - 4m - 6 = 0$
- 36. Solve: $x^2 + 5x + 2 = 0$

Graphing Quadratics

- 37. For the equation $y = x^2 - 4$, identify the vertex and the roots (x-intercepts).
- 38. For the equation $y = -x^2 + 6x - 5$, determine whether the parabola opens upward or downward.

Mixed Applications

- 39. A rectangular garden has dimensions $(x + 3)$ meters by $(x - 2)$ meters. Write a polynomial expression for the area, then expand it.
- 40. The height of a ball thrown upward is given by $h = -16t^2 + 64t + 5$. At what time(s) will the ball reach a height of 53 feet? (Solve $-16t^2 + 64t + 5 = 53$)